Michael Ragan

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Staff Software Engineer | TypeScript/Node.js, Distributed Systems & AI-Augmented Engineering

PROFESSIONAL SUMMARY

Staff-level Full-Stack Engineer with 17 years of experience building and scaling customer-facing software, APIs, and services across fintech, financial exchange, and enterprise SaaS platforms. Deep expertise in TypeScript/Node.js, PostgreSQL, and AWS cloud infrastructure, with a track record of owning systems end-to-end – from architecture through production – and shipping quickly without sacrificing reliability or security. Applies AI tooling throughout the development lifecycle to accelerate implementation, testing, and iteration, while retaining strong engineering judgment over what ships. Currently leading engineering for MEMX's real-time IPO auction platform, a U.S. stock exchange, where exchange-grade reliability, security, and speed of delivery are the daily bar.

CORE COMPETENCIES

TypeScript/Node.js Development Scalable APIs & Services Distributed Systems Exchange & Market Infrastructure AI-Augmented Development Cloud Architecture (AWS/Kubernetes) Test Automation (Playwright/Jest) End-to-End Ownership Technical Leadership

TECHNICAL SKILLS

Core Stack: TypeScript, Node.js/Express, React, Next.js, PostgreSQL, AWS

AI-Augmented Development: Claude, Cursor, GitHub Copilot

Frontend: React, TypeScript, Next.js, Redux/RTK, GraphQL

Testing: Playwright (E2E), Jest, TDD, component & integration testing, CI/CD gating

Backend: Node.js/Express, Java/Spring Boot

Cloud & Infrastructure: AWS (ECS, Lambda, RDS, S3, CloudFront, Cognito), Kubernetes, Docker, GitLab CI/CD

Data & Databases: PostgreSQL, Aurora, DynamoDB, Oracle SQL

Security & Identity: OAuth, SAML, JWT, RBAC, Okta

PROFESSIONAL EXPERIENCE

Lead Software Engineer — Oct 2025 – Present
MEMX — *Remote*

- Lead a 4-person team across all of MEMX's applications and own the IPO auction admin platform end to end – market infrastructure for booking, controlling, and monitoring live auctions in real time. Delivered in 4 months with one junior engineer alongside me – covered about 90% of the React/TypeScript frontend and the entire Node.js/Express backend
- Stream order book updates, auction metrics, and list state over WebSockets to all connected users simultaneously; sustained low-latency delivery under 500ms with multiple concurrent viewers per auction
- Built the test suite from scratch – 36 component tests, 8 Playwright E2E tests. Coverage sits around 90%, past the 80% bar we'd set
- Run three-tier RBAC on top of Okta groups, checked on every request; added an LRU cache so JWT validation wouldn't become a bottleneck
- Designed a RESTful mutation layer with last-write-wins concurrency to reliably handle block orders exceeding 100+ line items and multi-million-share volumes

Lead Full-Stack Engineer — Jan 2024 – Jul 2025
Convera — *Remote*

- Led an 8-person, frontend-focused pod building the microfrontend for the cross-border payments module, as part of the broader effort pulling payment functionality out of a monolith and into microservices. Shipped two weeks ahead of schedule, no major incidents after launch
- Co-led company-wide AI adoption alongside fellow tech leads – ran prompt-engineering training, introduced usage guardrails, and coached engineers on recognizing when a model was confidently wrong
- Used AI tools day-to-day to scaffold new components and features, generate test cases, and build reports for management, speeding up routine implementation without loosening code review standards

- Built the cross-border payments microfrontend alongside transaction dashboards, KYC onboarding, and sanctions screening – for a product with 100K+ daily users, deployed through S3 and CloudFront
- Implemented Jest and Playwright as mandatory merge gates in GitLab CI/CD, sustaining an 80%+ test coverage standard across an 8-person pod
- Added a GraphQL layer in front of the payment approval flow so the microfrontend could get real-time deltas without overfetching from the REST API underneath; implemented AWS Cognito-based role authorization across the payment and screening surfaces the pod owned
- Built an audit logging pipeline on DynamoDB to capture every user action at clickstream-level volume; DynamoDB Streams fed a consumer Lambda that normalized events into a queryable Postgres copy for compliance and support. Also used DynamoDB's low-latency reads/writes to power inline fraud and velocity checks (e.g., flagging accounts that created more than a threshold number of users within an hour)
- Peak load was hammering our Postgres/Aurora queries. Tuned the worst offenders and got throughput up about 45%

Principal Software Engineer Feb 2017 – Jan 2024
Oracle *Remote / Redwood Shores, CA*

- Rebuilt internal tools as React SPAs with Redux and Oracle JET, cutting support tickets for those tools by roughly half; later architected a real-time call monitoring dashboard using GraphQL and WebSocket subscriptions so managers got live call status and KPIs without waiting on a report
- Mentored 15+ engineers over the years, including on frontend structure and test coverage expectations; a good number of them moved into senior or management roles afterward
- Pushed several monolith-to-microservices efforts that let teams deploy on their own schedule instead of waiting on a shared release train
- Ran Kubernetes clusters on OCI handling 10M+ requests a day. Kept uptime at 99.99% through regular chaos testing
- Designed the data models and Java/Spring backends for datasets north of 500GB. Got the slow paths about 70% faster

Senior Software Engineer Jan 2013 – Feb 2017
Oracle

- Built distributed backend services with Java, Spring, and Hibernate; used Maven for build consistency across 20+ microservices
- Developed JavaScript libraries and PHP services with MySQL that let customers embed our applications; over 50 enterprise customers used this
- Built SSO authentication in PHP and Perl that cut login issues by about 60%

Software Engineer Oct 2008 – Dec 2012
Oracle

- Moved from technical support into a software engineering role, building production code within the first year
- Created JavaScript and Oracle JET frontends that improved user engagement by about 35%
- Did Linux system admin work and improved CI/CD with Jenkins and Docker; cut build times by more than half

EDUCATION

B.S. Computer Science (Network and Communications) March 2007
DeVry University, Fort Washington, PA